Baowen Zhang

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EDUCATION

SUNY College of Environmental Science and Forestry **Aug 2020 – Jun 2026**
Ph.D. in Sustainable Construction Management & Wood Science – GPA: 3.60/4.00 *Syracuse, NY*
— Advisor: Prof. Paul Crovella. Research in mass timber, A5 construction-phase LCA, ML-based retrofit forecasting, and IFC digital twin energy modeling.

Stevens Institute of Technology **Aug 2018 – Dec 2019**
M.Eng. in Engineering Management – GPA: 3.78/4.00 *Hoboken, NJ*
— Coursework: Project Management, Engineering Economics, Risk Analysis, Lean Six Sigma.

Tianjin University of Technology **Sep 2014 – Jun 2018**
B.Eng. in Construction Engineering – GPA: 3.4/4.00 *Tianjin, China*
— Coursework: Construction Estimating, Structural Mechanics, CAD Drafting, BIM Applications, Cost Engineering, Project Management.

TEACHING EXPERIENCE

Visiting Assistant Professor **Jan 2026 – Present**
Colorado State University Pueblo *Pueblo, CO*
— Serve as Instructor of Record for undergraduate courses: *Construction Management (CET 303)*, *Construction Engineering (CE 361)*, *Heavy/Highway Cost Estimating (CET 305)*, *Wood Structural Systems (CM 330)*, and *Mechanical and Electrical Systems (CM 451)*.
— Develop and deliver instruction in construction estimating, engineering economics, CPM scheduling, earned value analysis, project delivery methods, and construction safety and law. Received an average student teaching evaluation of 4.6/5.0.
— Organize, mentor, and lead the CSU Pueblo student team for the ASC Regions 6 & 7 Student Competition, coordinating team preparation, technical training, and competition participation.
— Contribute to ABET accreditation preparation for the Civil Engineering program through course assessment, collection and evaluation of student work samples, and documentation of student learning outcomes aligned with ABET Engineering Accreditation Commission (EAC) criteria.

Graduate Teaching Assistant **Aug 2020 – Dec 2025**
SUNY–ESF *Syracuse, NY*
— Assisted instruction for undergraduate courses including Construction Materials, Revit/BIM, Estimating, Scheduling, Engineering Economy, Applied Structures, and Sustainable Buildings in in-person and hybrid formats.
— Supported laboratory, field, and project-based learning activities, including surveying demonstrations, BIM modeling, and construction planning exercises.
— Designed assignments and workshops to strengthen student learning in cost estimating, scheduling, and building performance analysis.
— Mentored 30+ students per semester through office hours, problem-solving sessions, and support for technical writing and course projects.
— Developed an AI-based grading and feedback system that reduced turnaround time by 20% while maintaining rubric consistency.

RESEARCH & PROJECTS

LLM-Agent Workflow for Building Energy Modeling **Aug 2024 – Present**
— Developing an MCP-enabled LLM-agent workflow to automate BIM-to-BEM model generation, enrichment, and energy simulation.
— Integrating IFC, Rhino/Grasshopper, Honeybee, and EnergyPlus to establish interoperable, simulation-ready building energy models.
— Incorporating thermal imagery and UAV point clouds to identify thermal bridging, envelope degradation, and geometric discrepancies.
— Developing automated model calibration and performance assessment workflows to support building energy audits and retrofit decisions.

NYS Buy Clean Concrete – Phase 3 GWP Limit Analysis **Jun 2026 – Present**
— Collaborating with the New York State Office of General Services (NYS OGS) to support embodied-carbon policy revisions under State Finance Law §136-d.
— Analyzed 32 third-party-verified concrete EPDs from nine producers, evaluating cradle-to-gate (A1–A3) GWP across strength classes, producers, and supply channels.
— Developed compliance and supply-feasibility scenarios for 10–25% GWP limit reductions; identified producer-level effects accounting for approximately 80–95% of observed GWP variation.
— Evaluating plant-level emissions, producer-specific EPDs, and procurement-volume-weighted performance to inform feasible low-carbon concrete procurement limits.

Sustainable Concrete with Low-Carbon SCMs **Jan 2025 – Nov 2025**
— Investigated sustainable concrete mixtures incorporating up to 80% cement replacement with ground granulated blast-furnace slag (GGBS).
— Conducted mechanical and durability testing, cradle-to-gate LCA, and multicriteria sustainability assessment.
— Published findings in *Cleaner Engineering and Technology* (2026).

Hardwood CLT Development and Testing **Jan 2024 – Nov 2024**

— Developed PRG-320-compliant cross-laminated timber (CLT) panels in collaboration with Michigan State University, utilizing over 4,000 board feet of urban hardwood.
— Conducted flexural, shear, and delamination testing, demonstrating 345% strength and 112% stiffness improvements.
— Advanced urban wood reuse and presented research findings to ASC technical committees.

Machine Learning for Building Energy Retrofit (NYC LL97) **2023 – 2024**

— Analyzed over 6,000 NYC Local Law 87 energy audits to evaluate retrofit strategies and compliance with Local Law 97.
— Developed ensemble machine learning models for post-retrofit energy prediction, achieving a 23% reduction in RMSE.
— Integrated energy savings, retrofit costs, and LL97 penalties into an economic decision-support framework; published in *Journal of Building Engineering* (2026).

Lignin-Based Insulation Material Development **2023 – 2024**

— Developed bio-based polyurethane insulation foams incorporating lignin and castor-oil-derived polyols.
— Evaluated thermal, mechanical, and material performance to assess potential applications in sustainable building insulation.
— Co-authored related publications in *Chemical Engineering Journal* (2024, 2025).

PUBLICATIONS

— **Zhang, B.** & Crovella, P. (2026). "An integrated machine learning approach to economically prioritize energy conservation measures for multifamily buildings under New York City's Local Law 97," *Journal of Building Engineering*, Vol. 128 (Article 116461). DOI
— **Zhang, B.**, Gu, H., & Crovella, P. (2026). "Quantifying Construction-Stage (A5) Energy Consumption in Mass Timber Buildings: A Dual-Track Assessment Framework," *Case Studies in Construction Materials*, Vol. 25 (Article e06280). DOI
Featured in USDA Forest Service's Rooted in Research newsletter(October 2026).
— **Zhang, B.**, Aman, M. I., Crovella, P., & Hossain, M. U. (2026). "Designing sustainable concrete to maximize the use of local waste materials: Mechanical, durability, and sustainability-based multicriteria analysis," *Cleaner Engineering and Technology* (Article 101335). DOI
— Zhang, B., Song, F., Sun, X., **Zhang, B.***, "Numerical Study on the Seismic Performance of Replaceable Energy-Dissipation Channel Steel Joints," *Advanced Steel Construction*, 2025 (Accepted for publication).
— Liang, W., Lu, J., Fu, J., Zhang, B., **Zhang, B.**, Wang, Y. (2025). "Analytical and Experimental Study on Bond Behavior of Embedded Through-Section FRP Bar-to-Concrete Joints Using a Trilinear Cohesive Material Law," *Buildings*, Vol. 16, No. 1 (Article 164). DOI
— Jeong, S., Zhang, M., Meng, X., Wang, Y., Scheidemantle, B., **Zhang, B.**, Min, H., Ji, A., Crovella, P., Cai, C. M., Ragauskas, A. J., Yoo, C. G. (2025). "Properties and performance of lignin-based polyurethane foams from lignin and castor oil as synergistic bio-polyols," *Chemical Engineering Journal*, Vol. 520 (Article 166159). DOI
— Zhang, M., Jeong, S., Cho, W., Ryu, J., **Zhang, B.**, Crovella, P., Ragauskas, A. J., Wie, J. J., Yoo, C. G. (2024). "Green co-solvent-assisted one-pot synthesis of high-performance flexible lignin polyurethane foam," *Chemical Engineering Journal*, Vol. 499 (Article 156142). DOI
— Wang, E., Lawal, A., Wu, W., Onyango, M. A., Wu, D., **Zhang, B.** (2022). "An Integrated Analytic Hierarchy Process: Markov Model for Rating Condition and Predicting Service Life of Retaining Walls," *Transportation Research Record*, Vol. 2676, No. 12.

* Corresponding author.

CONFERENCE PAPERS AND PRESENTATIONS

— **"Quantifying Construction-Stage (A5) Energy Consumption in Mass Timber Buildings"** Invited Presentation, North America Wood LCA Group Quarterly Meeting, June 2026.
— **"Human-in-the-Loop, Not Hands-Off: Do AI-Assisted TA Comments Improve Construction Writing Quality and Fairness in an Introductory BIM Course?"** Conference Paper, Associated Schools of Construction (ASC) 62nd Annual International Conference, 2026. Published in *EPiC Series in Built Environment*, Vol. 7, pp. 226–235. DOI: 10.29007/278n.
— **"Hardwood CLT Performance and A5 Construction-Phase Carbon Assessment"** Invited Speaker, FPS-SWST International Conference, Fort Collins, CO, 2025.
— **"Machine Learning-Based Energy Retrofit Forecasting for NYC Local Law 97 Compliance"** Oral Presentation, Associated Schools of Construction (ASC) Annual Conference, 2023.

ACADEMIC SERVICE & MEMBERSHIPS

— Reviewer, *Transportation Research Record (TRR).*
— Reviewer, *Buildings.*
— Reviewer, Associated Schools of Construction (ASC).
— Member, U.S. Green Building Council (USGBC).
— Member, Urban Green Council.

CERTIFICATIONS

— OSHA 10-Hour Construction Safety and Health (2025).